

Saaras Pakanati

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Education

Rice University, PhD Mechanical Engineering (Student) Advisor: Prof. Daniel Preston	August 2026 – Present GPA –/4.00
University of Cincinnati, BS Mechanical Engineering, <i>Summa Cum Laude</i> <i>Minor</i> in Mathematics Advisor: Prof. Kishan Bellur	August 2021 – May 2026 GPA 3.96/4.00

Professional Experience

Undergraduate Research Assistant University of Cincinnati Lab for Interfacial Dynamics, Cincinnati, Ohio - Numerical & Experimental study of multiphase interfaces and associated phenomenon.	August 2024 – May 2026
New Product Development Mechanical Engineering co-op MSA Safety, Cranberry Township, Pennsylvania - Used Ansys FEA to validate safety device designs and Minitab-based DOE to optimize TRL1 gas detection prototype testing.	January 2024 – April 2024
Engineering co-op (2 Semesters) Richards Industrials, Cincinnati, Ohio - Built Visual Basic tools for process optimization and quality analysis, supported by hands-on valve testing and data analysis.	August 2022 – August 2023
Software Intern Visual Collaboration Technologies, Inc., Bengaluru, Karnataka, India - Built a Python Dash visualization tool for a CFD study while following Agile development and PEP8 coding standards.	June 2022 – July 2022

Awards, Fellowships, and Honors

Dean's Prize – \$10,000 - Rice University, Office of Graduate and Postdoctoral Studies	2026
Technical Excellence in Senior Design Project: 2nd place – \$1,000 - University of Cincinnati, College of Engineering and Applied Science	2026
Merchant, M. Eugene Scholarship – \$1,500 - University of Cincinnati, Mechanical & Materials Engineering Department	2026
Armstrong Discovery Fellowship (2×) – \$11,000 (total) - Armstrong Institute for Space, Technology, and Research	2025-2026
CEAS Annual Scholar: Undergraduate Research – \$1,000 - University of Cincinnati, College of Engineering and Applied Science	2025
Undergraduate Research co-op Fellowship (2×) – \$18,000 (total) - University of Cincinnati, College of Engineering and Applied Science	2024-2025
UC Calculus Contest: 2nd place – \$250 - Cengage Publishing & UC Department of Mathematical Sciences	2022
CEAS International Outreach Scholarship – \$40,000 (total) - University of Cincinnati, College of Engineering and Applied Science	2021-2026
University of Cincinnati Global Scholarship – \$8,000 (total) - University of Cincinnati	2021-2026

Journal Publications

[†] denotes equal contribution

- J2. A. Sarchami, **S. Pakanati**, A. Yasin, M. Jog, K. Bellur, *An experimental study of interfacial dynamics control using temperature-sensitive surfactants*, Langmuir, 2026
10.1021/acs.langmuir.5c06748
- J1. A. Yasin[†], **S. Pakanati**[†], K. Bellur, *A multiscale CFD model of evaporating Hydrogen menisci: Incorporating subgrid thin-film dynamics and in situ accommodation coefficients*, Fuels, 2026
10.3390/fuels7010003

Conference Presentations

* denotes presenting author

- C2. A. Sarchami, **S. Pakanati**, T. Enam, A. Yasin, K. Bellur*, *Stability and Contact Line Dynamics of Evaporating Thin Liquid Films*, 11th ASTFE Thermal and Fluids Engineering Conference, Tempe, Arizona, March 9–12, 2026
- C1. A. Sarchami*, **S. Pakanati**, K. Bellur, *Dynamic Optical Measurement of Evaporating Liquid Films*, ASME International Mechanical Engineering Congress & Exposition, Memphis, Tennessee, November 16–20, 2025 (Poster)

Invited Talks

- I1. *A Multiscale Approach to Modeling Evaporation in Cryogenic Space Fuels* July 2026
Mechanical, Materials & Industrial Engineering Department Student Seminar, University of Cincinnati, Cincinnati, Ohio

Teaching

- Grader** August 2025 – December 2025
Dept. of Mechanical & Materials Engineering, University of Cincinnati, Cincinnati, Ohio
- Graded assignments and proctored exams for 150+ students in MECH 2010 (Thermodynamics).
- Supplemental Review Session Instructor** January 2023 – May 2025
University of Cincinnati Learning Commons, Cincinnati, Ohio
- Led review sessions for Calculus I & II and peer tutored across 15 Engineering, Mathematics, Physics, and Chemistry courses.

Other Involvements

- University of Cincinnati's Undergraduate Research Society** September 2021 – May 2026
- President (2025–26), Vice President (2024–25) and Events Chair (2022–23 & 2023–24)
- Advocated for undergraduate research funding while promoting research participation among underrepresented students.
- Bearcat Buddies** October 2021 – December 2023
- Volunteered weekly to tutor K-12 students in the Cincinnati Public School system.

Outreach Talks

- O2. *Undergraduate Research: What, Why, How, and Co-ops*, University of Cincinnati, Cincinnati, Ohio 2024–2026
- Recurring presentation encouraging first-year engineering students to pursue undergraduate research.
- O1. *Undergraduate Research*, UC Futures, University of Cincinnati, Cincinnati, Ohio 2025
- Encouraging high school students from rural and small-town Ohio to pursue research opportunities.

Skills

- Software:** Ansys & Siemens simulation suites, OpenFOAM, ParaView, SolidWorks, LAMMPS, OVITO, Inkscape
Programming: Python, MATLAB, C, C++, Bash, R Language, Minitab, LabView, L^AT_EX
Experimental: Optical Interferometry, Basic Machining, High Speed Imaging

Certifications & Licenses

- OASIS Foundations Program**, *University of Cincinnati* – 2025 | [link to certificate](#)
MATLAB Fundamentals, *Mathworks* – 2023 | [link to certificate](#)
IBM Data Science Practitioner, *IBM* – 2022 | [link to certificate](#)
Six Sigma Yellow Belt, *The Council for Six Sigma Certification* – 2022 | Credential ID: h7L0bACNv5
NX – CAD Design Associate, *Siemens* – 2022 | Credential ID: 6763-164-884-2231